

Business Requirement Document

E-Commerce Order Management Database System — Week 1 Deliverable

1. Introduction

This document defines the business requirements for the E-Commerce Order Management Database System. It expands on the Requirement Analysis Report by stating precisely what the system must do (functional requirements), the quality attributes it must exhibit (non-functional requirements), what is and is not included in the current project phase (scope), and the assumptions and constraints under which the analysis was carried out.

2. Functional Requirements

2.1 Customer Management

- 1 The system shall allow a new customer to register with name, email, mobile number, address, and registration date.
- 2 The system shall prevent duplicate registration using the same email address.
- 3 The system shall allow a customer's contact details and address to be updated.
- 4 The system shall allow retrieval of a customer's complete order history.

2.2 Category & Product Management

- 1 The system shall allow creation, update, and removal of product categories with a name and description.
- 2 The system shall allow a product to be added with name, price, stock quantity, and an associated category.
- 3 The system shall allow product price and stock quantity to be updated.
- 4 The system shall allow products to be searched and filtered by category.

2.3 Supplier Management

- 1 The system shall maintain supplier name and contact information for procurement purposes.
- 2 The system shall associate suppliers with the products they provide.

2.4 Inventory Management

- 1 The system shall reduce a product's stock quantity automatically when an order is confirmed.
- 2 The system shall prevent an order from being placed for a quantity exceeding available stock.
- 3 The system shall allow stock quantity to be increased when new stock is received from a supplier.

2.5 Order & Order Details Management

- 1 The system shall allow a customer to place an order containing one or more products.
- 2 The system shall record the order date, total amount, and order status for every order.

- 3 The system shall record, for every product in an order, the quantity ordered and the unit price at the time of purchase in the Order Details entity.
- 4 The system shall allow the order status to progress through defined states, such as Pending, Confirmed, Shipped, Delivered, and Cancelled.
- 5 The system shall automatically compute the total amount of an order from its associated order detail lines.

2.6 Payment Management

- 1 The system shall allow a payment record to be created against an order, capturing payment method, payment date, and payment status.
- 2 The system shall support multiple payment methods (e.g. Credit/Debit Card, Net Banking, Cash on Delivery, Digital Wallet).
- 3 The system shall not allow an order to proceed to shipment unless its payment status is confirmed as successful, except where Cash on Delivery is selected.

2.7 Shipment Management

- 1 The system shall create a shipment record for a confirmed order, capturing the delivery address, shipment date, and delivery status.
- 2 The system shall allow delivery status to be updated as the shipment progresses (e.g. Dispatched, In Transit, Delivered, Returned).

2.8 Review Management

- 1 The system shall allow a customer to submit a rating and comment for a product.
- 2 The system shall allow retrieval of all reviews for a given product to display an average rating.

2.9 Reporting

- 1 The system shall support generation of reports such as sales by product/category, revenue over time, top-rated products, and order fulfilment/delivery performance.

3. Non-Functional Requirements

Attribute	Requirement
Performance	The system shall retrieve product catalogue and order history results within acceptable response times even as the number of orders and products grows into the thousands.
Scalability	The database design shall accommodate continuous growth in customers, products, and orders without requiring structural redesign.
Data Integrity	The system shall enforce referential integrity between related entities (e.g. an Order Detail must reference a valid Order and a valid Product) and shall avoid unnecessary data redundancy through normalization.
Security	Customer personal data and payment information shall be protected from unauthorized access, and access to administrative operations shall be restricted to authorized roles.
Reliability / Availability	The system shall be available for order placement and processing with minimal downtime, given the continuous nature of online shopping.
Usability	Data entry and retrieval operations shall be structured so that catalogue, order, and reporting staff can use the system with minimal training.
Maintainability	The database schema shall be documented and organized so that future changes (e.g. new attributes, new report types) can be made with minimal impact on existing functionality.
Auditability	Key events (order status changes, payment status changes, delivery status changes) shall be traceable through recorded dates and statuses.
Backup & Recovery	The system shall support regular backup of the database to prevent loss of business-critical data.

4. Project Scope

4.1 In Scope

- Design and implementation of a relational database covering the nine core entities: Customer, Category, Product, Supplier, Order, Order Details, Payment, Shipment, and Review.
- Entity-Relationship modelling and normalization of the above entities.
- SQL implementation of tables, relationships, and constraints derived from this analysis.
- Generation of business reports based on data captured by the core entities (e.g. sales, revenue, delivery performance, product ratings).
- Core business workflows: registration, catalogue browsing, order placement, payment, shipment tracking, and reviews.

4.2 Out of Scope

- Front-end / user-interface design and development for customers or administrators.
- Integration with real third-party payment gateways or shipment/courier APIs.

- Marketing features such as discount coupons, loyalty programs, or recommendation engines.
- Multi-currency or multi-language support.
- Mobile application development.

5. Assumptions and Constraints

5.1 Assumptions

- Each customer registers with a unique, valid email address used as their primary contact.
- A product belongs to exactly one category at a time.
- An order can contain multiple products, and a product can appear in multiple orders, resolved through the Order Details entity.
- Each order is associated with exactly one payment record and, once confirmed, exactly one shipment record.
- A customer may submit a review only for products relevant to their purchase history.
- Stock levels are managed solely through the Product entity's stock quantity attribute at this stage of the project.

5.2 Constraints

- Only the predefined nine core entities and their listed attributes may be used; no entities or attributes may be added, removed, or renamed, per project instructions.
- The project timeline follows a weekly milestone structure, limiting this phase to requirement analysis and documentation without database implementation.
- The system is being designed as an academic semester project and therefore does not integrate with live external systems (payment gateways, courier services, etc.).
- All documentation must be original work prepared by the student.